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EAGLE STREAM ODMS CRD SURVEY REPORT (SERVER SEGMENT)

Customer participated :

Inventec (SH & TW), Wistron ZS, Quanta (SH & TW), Foxconn TJ, Pegatron TW

20ww32 Rev 1

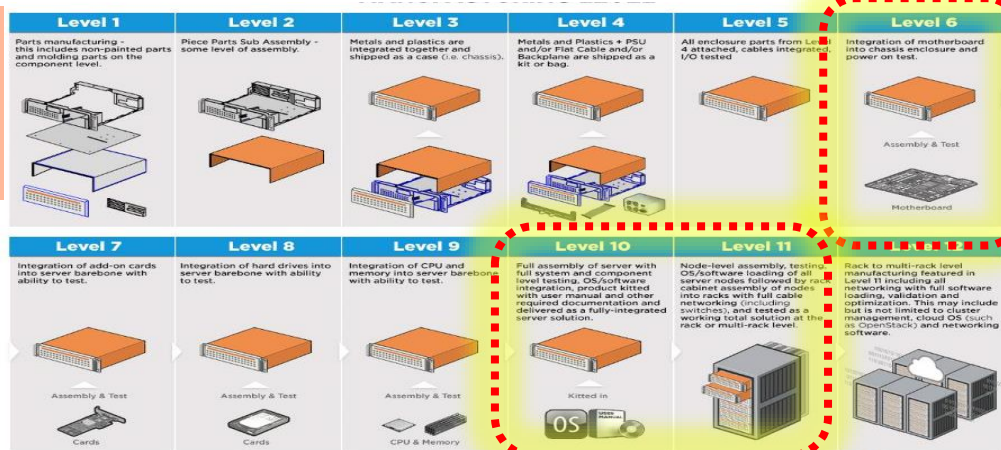
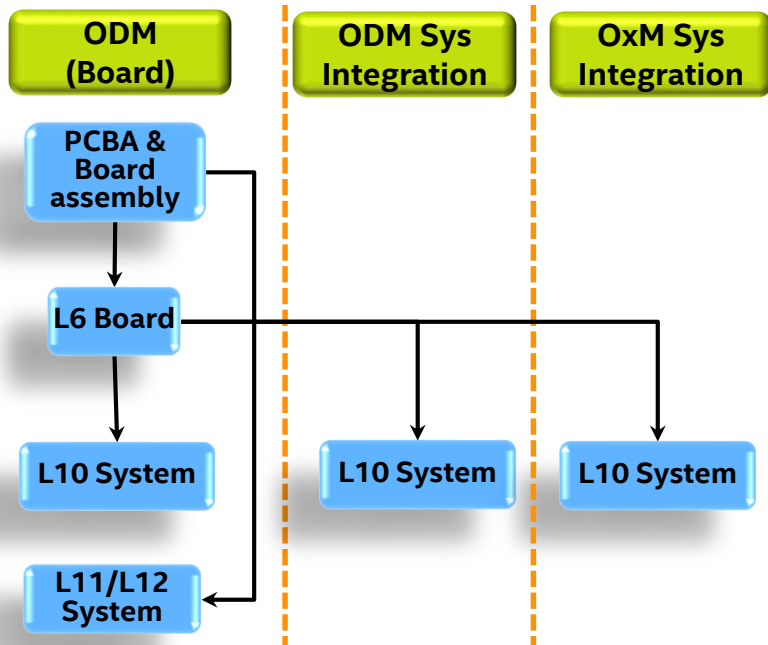
Agenda

- ❖ Server Customer Manufacturing
- ❖ PCBA Material / Consumables
 - Technology & Process
- ❖ PCBA Machine / Equipment
 - Process Parameter & Capability
 - Paste Print, PnP, Reflow, Wave Solder, Rework, Inspection (SPI, AOI)
- ❖ Others & Automation
 - PHM Assembly, Manufacturing Test, System Integration
- ❖ Key Takeaway
- ❖ Next Step

Server Manufacturing Level & Customer Landscape

Enterprise Customer : Dell, HP, Lenovo

CSP Customer : FB, Amazon, MSFT, Google



Level	Description
L6	Integration of motherboard into chassis enclosure and power on test.
L10	Full assembly of server with full system and component level testing, OS/software integration, product kitted with user manual and other required. Documentation and delivered as a fully-integrated server solution.
L11	Node-level assembly, testing, OS/software loading of all server nodes followed by rack cabinet assembly of nodes into racks with full cable networking (including switches), and tested as a working total solution at the rack or multirack level.

PCBA Material / Consumables

Data Source:
Server ODMs

Material	Inventec	Quanta	Foxconn TJ	Wistron ZS	Pegatron TW
Stencil	Local supplier	Local supplier	Foxconn; 深精藝-SHENJINGYI; 蒙瑞-MENG RUI	Local supplier	Iden Win-laser
PCB	Local supplier	Gold Circuit, Tripod, ISU	GCE; BTK; Tripod	Hannstar, Boardtek, GCE	ACCL, GCE, Compeq
Solder Paste (SMT)	Shenmao PF606-P	Shenmao PF606-P Shenmao PF606-P245	ALPHA OM338PT Senju M705-S101HF-S4	Shenmao PF606-F Henkel GC-10	Shenmao PF606-P SMIC M705-S101HF(N6)-S5
Rework	Not provided	SRT/Summit 1800	Conceptronic/ Feedom2000	SRT 1100LX, 2200LX	SUMMIT-Lxi
Paste/Flux (Rework)	Not provided	Shenmao PF606-P Shenmao PF606-P245	Alpha OM338PT	Shenmao SMF-2	Not provided
Flux (Wave solder)	Shenmao SM816	Shenmao SM816	Not provided	Shenmao SM816, Alphametal EF-6808HF	Shenmao SM-827

Key Message :

- Prefer Stencil local vendors for fast response, delivery time & on-site support.
- Wave solder process is still required for thru-hole components.
- Shenmao paste/flux is commonly used except Foxconn.
 - Shenmao paste is generally used during early development; will be used as back-up during mass production (MP).
 - Alternate solder paste is mandatory for SMT per their customer requirement.

PCBA Material / Consumables & Technology

Data Source:
Server ODMs

Material	Technology	Inventec	Quanta	Foxconn TJ	Wistron ZS	Pegatron TW
Stencil	Cut / Coating	laser cut + nano coat	Laser cut	electroform, laser cut with electro polish + nano coat	Laser cut with electro polished	Laser cut with etching
	Thickness (mils)	4 & 5	5	4 (typical)	5 (per MAS)	5
	Step up/down (mils)	Yes, 4, 6 & 8	Yes, 4 - 6	Yes, 3, 5, 6, 7, 8 & 10	Yes, +1	Yes, 4 - 6
PCB	Surface Finished	OSP	ENIG, OSP, ImAu & etc	OSP	OSP (majority), ENIG (few)	OSP
	Thickness (mils)	Not provided	62, 79, 98 & 118	62 & 98	62, 98, & 118	79
	Type	Not Provided	Type 4	Not Provided	Not Provided	Type 4

Key Message :

- Stepped stencil is very common for server ODMs, due to board design complexity (eg, large socket/ connectors and small passive components). Stepped up/down range between 3 – 10 mils.
- Stencil thickness for Intel component 4 – 5 mils .
- PCB thickness is within TMDG guideline.

PCBA Machine (Brand & Model)

Data Source:
Server ODMs

Area	Inventec	Quanta	Foxconn TJ	Wistron ZS	Pegatron TW
Paste Print	ASM DEK/horizon 03IX	Fuji/GPX-C	ASM DEK/DEK 03i	ASM DEK/ <i>Infinity API; Horizon; Neo Horizon 03iX</i>	DEK/ Neo Horizon 03iX
PnP	Fuji/NXT II	Fuji/AIMEXIII, NXTIII, NXTII	Universal/GSM, Genesisa Panasonic/CM602&CM 402	Panasonic/CM602F0,DT 401,NPM	Fuji/NXT M6*14
Reflow	<i>REHM</i>	REHM/VXP-945 ERSA/Hotflow 3/26 XL	Vitronics/XPM ² 1240	REHM/ <i>VXP-944, VXP945 (N)</i>	Heller/2043 MK5
Rework	Not provided	SRT/Summit 1800	Conceptronic/ Feedom2000	SRT 1100LX, 2200LX	SUMMIT-Lxi
Wave Solder	Speed line/ electra	Ersa Hotflow PowerFlow N2	Not provided	JT Smart610-H, ChanLong/600CNP-HA	JYI DIANN JT-620L
SPI	TRI-7007L	Parmi Sigma XXL, SAKI BF-3Si (3D SPI), Cyber Optic SE5000L	Not provided	TRI-7007L	TRI-7007 SII
SMT X-ray	VTROX	VTROX V810 XXL	Not provided	Phynix-2D Xray, TRI TR7500L	Nordson DAGE JADE FP
Auto Inspection	TRI-7700	SAKI BF10Z, (S/S, C/S) RE-3Di-7S2 (Post W/S)	Not provided	Not provided	Not provided

Key Message :

- Wide variety of machine in server ODMs. More challenges in equipment capability assessment & development for future package design.

PCBA Process Parameter/Capability – Paste Printing

Data Source:
Server ODMs

Parameter / Capability	Inventec	Quanta	Foxconn TJ	Wistron ZS	Pegatron IW
Squeegee Speed (mm/s)	20 - 60	1 - 200	45	40	40
Squeegee Pressure (Kg)	8-15	1 – 25.5 (automatic control)	10	6 - 9	11 - 14
Stencil Separation Speed (mm/s)	0.2-1	5.5	0.5	0.2~0.8	0-1
Under Screen Cleaning frequency (per print)	1~3	1 (Top Side)	Bottom side : 3 Top side : 2	1~4	Bottom side : 4 Top side : 2
Stencil Cleaning Frequency (hrs)	12	2	12	6	Not provided
Cleaning Process	Ultrasonic	Ultrasonic	The machine automatically to clean	Ultrasonic	Ultrasonic
PCB support type	Vacuum & clip	Panel Vacuum	support block	Pin or Block	Vacuum and pin
Do you use the same pallet for paste print, PnP and reflow?	No	Yes, if use pallet for paste print (case by case)	Yes, if use pallet for paste print (case by case)	No	By model
Maximum board handling (inch, length x width)	24 x 20	24 x 24	24 x 20	24 x 20	24 x 20

Key Message :

- Varies PCB support type used at paste print. Pallet use depend on case/model.
- PCB maximum size > 20 inch (width) may require tool upgrade for most customers.

PCBA Process Parameter/Capability – PnP

Data Source:
Server ODMs

Parameter / Capability	Inventec	Quanta	Foxconn TJ		Wistron ZS	Pegatron TW
Nozzle size / ϕ (mm) (LGA socket & Intel BGA)	20, 15	H08M (45 x 45) H01/H02 (74 x 74)	8.65, (Universal 340F)		15, (#5401 Nozzle)	20
Max BGA/socket size capability (mm, x & y)	X:74 Y:74	X:74 Y:74	BGA X: 55 Y: 55	Socket X: 83 Y: 64	X: 100 Y: 80	X:74 Y:74
Max BGA/socket weight capability (g)	< 50	< 90	< 28.5		< 100	< 38.5
Using pallet during placement	No	Board type dependent (<1.6 mm thickness)	Yes, if use pallet for paste print (case by case)		No	By model (pin support)
Maximum board handling (inch, length x width)	24 x 21	30 x 30	24 x 20		26 x 18	21 x 16
min pitch/ball recognition capability	0.4 pitch/0.25 bump	Lead: 0.24mm Bump: 0.40mm Pin: 0.40mm	0.4 pitch		0.3 mm	0.4 mm

Key Message :

- LGA/BGA beyond socket-E had reached max weight capability for some customer, need early development roadmap & engagement.
- PCB maximum size > 16/18 inch (width) require tool upgrade for Pegatron/Wistron.
- PnP ball/lead recognition capability > 0.4 mm pitch.

PCBA Process Parameter/Capability – Reflow

Data Source:
Server ODMs
Pegatron, Foxconn, Inventec

Parameter/ Capability	Inventec	Quanta	Foxconn TJ	Wistron ZS	Pegatron TW
Use of PCB support	Carrier	Bottom : Center support Top : Alloy Carrier	Main board usually use the carrier to support PCB.	Not provided	Special carrier
N2 purging	Yes	Yes	Yes	Yes	Yes
O2 PPM control	< 3000	< 3000	< 1000	< 1000	< 2000
Air flow control	Yes	Yes, Left to Right (facing the machine)	Yes	No	Not provided
Belt Speed (mm/min)	900	900	250 -2030	900	850-1000
Ramp Rate (°C/s)	≤ 3	≤ 3	≤ 3	≤ 3	≤ 3
Cooling Rate (°C/s)	≤ 3	≤ 3	≤ 3	≤ 3	≤ 3
Peak Temp Range	230-250 °C	230-250 °C	235-255 °C	230-250 °C	230-250 °C
Time above Liquidus (s)	60 - 90	60 - 90	60 - 120	60 - 90	60 - 120
Soak Temp Range (s)	90 -120	90 - 120	60 – 120 (145 – 175°C)	90 - 120	60 - 120
Delta T across Intel component	15°C	10°C	5°C	10°C	10°C
Number of reflow and cooling zones	8- ramp,3- reflow,3- cooling	REHM:9/5 ERSA: 9/3	12- reflow, 4- cooling	9- ramp, 4- reflow, 5- cooling	13 heating, 3 cooling

Key Message :

- Customer's NPI typically followed MAS recommended profile. They customize mass production (MP) profile in order to achieve good yield.
 - Peak Temp range of 20 °C due to large components on PCB. Foxconn required range 235-255 °C, others @230-250 °C
 - TAL range 60 – 90s for most customer, Foxconn & Pegatron required range 60 -120s.
 - Inventec is Delta_T <15°C (vs MAS <10°C).

PCBA Process Parameter/Capability – Wave & Rework

Data Source:
Server ODMs

Area	Parameter	Inventec	Quanta	Foxconn TJ	Wistron ZS	Pegatron TW
Wave soldering	Preheat Time (s)	<180	Not provided	192 - 240	by product	90 - 130
	Solder Pot Temperature (°C)	265 - 275	265 - 275	260 - 270	263 - 270	270
	Conveyor Speed (mm/min)	900	500 - 2500	600 - 750	600 - 800	850
	Components / pitch undergo wave solder	connector, dip capacitor	TH component, >2.5mm pitch	Not provided	E-Cap, DIMM/IO Connectors, etc... > 1.57mm pitch	All PTH component (DIP type)
	O2 PPM monitor	Not provided	<3000 O2 PPM	<1000 O2 PPM	NA	NA
Rework	Peak Temp Range (°C)	235 - 250	50 - 400 (typo?)	235 - 255	230 - 250	230 - 250
	Time above Liquidus (s)	30 - 120	60 - 90	60 - 120	60 - 90	Socket: 60 - 120 BGA: 60 - 90
	Soak Temp Range and Time	90 - 120	90 - 120	60 - 120 (145-175 °C)	90 - 120	Socket: 60 - 100 BGA: 60 - 120
	Max BGA/socket weight capability (grams)	Not provided	< 900	< 28.5	< 100	< 38.5
	Is N2 used& O2 PPM control ?	Not provided	Yes, <3000 O2 PPM	No, no O2 PPM control	Yes, < 3000 O2 PPM	Not provided

Key Message :

- Rework PnP had reached maximum weight capability for Foxconn & Pegatron (<28g).
- Rework @ air is still required.
- Peak Temp range of 20 °C required to achieve good yield (same as SMT).

PCBA Process Parameter/Capability – Inspection (1)

Data Source:
Server ODMs

Area	Process	Inventec	Quanta	Foxconn TJ	Wistron ZS	Pegatron TW
SPI	Sampling rate	100%	100%	100%	100%	100%
	Min/Max Spec and control limit	not provided	not provided	not provided	not provided	not provided
	Is SPI Performed on selective components or 100%?	100%	100%	100%	100%	Yes, selective
	Maximum board handling capability (inch, length x width)	26 x 24	32 x 24	24 x 20	26 x 24	24 x 20
X-Ray	In line / offline	offline	In-line	offline	AXI for In-line, 2DX-ray for offline	offline
	sampling rate	Per customer requirement	NPI: 100% MP: Selective (BGA, Socket, TH component)	Sample 5 pcs/lot start, follow by 1 pcs per hour	Not provided	NPI: 100% MP: 1 pcs every 2 hrs

Key Message :

- SPI is performed 100% for every PCB.
- Post SMT X-ray sampling plan varies by customer due to different run rate & line set-up.

PCBA Process Parameter/Capability – Inspection (2)

Data Source:
Server ODMs

Area	Process	Inventec	Quanta	Foxconn TJ	Wistron ZS	Pegatron TW
AOI	In line / offline ?	In-line	In-line	In-line	In-line	Before in line / After off line
	sampling rate ?	100%	100%	100%	100% for production	100% for production
	How many AOI's are used?	2	4	2	Not provided	4
	Where is AOI located?	Post Reflow	Pre-reflow: 2 (CS/SS)*, Post wave solder: 2 (CS/SS)	Post Reflow	Post reflow	Pre-reflow: 2, Post-reflow: 2
	Is AOI Performed on selective components or 100%?	100%	NPI: 100% MP: Selective	100%	100%	Yes (no detail provided)
	Maximum board handling capability (inch, length x width)	26 x 24	34 x 27	24 X 20	26 x 24	24 X 20

*C/S – Chip Shooter; S/S – Socket Shooter

Key Message :

- 100% post-reflow AOI is performed during MP phase by in-line process for all customers.
- Only 2 customers performed pre-reflow AOI for early issue detection.

PCBA – Operation & Others

Data Source:
Server ODMs

Area		Inventec	Quanta	Foxconn TJ	Wistron ZS	Pegatron TW
2D Scanner	Purpose	inventory check, traceability	shop floor scanning for traceability	inventory check, traceability	inventory check, traceability	Traceability
	Which process	kitting,SMT, etc.	SMT & System Final assembly	All process	IQC, SMT, Assembly/Testing process, Rework process	SMT
	In line or offline	In line	In-line assembly (SMT/System Assy)	In line traceability	Both In line and offline	In-line
Adhesive	Using adhesive for BGAs	no	no	no	no	Yes, *not on Intel part
	Type of adhesive used	N/A	N/A	N/A	N/A	Not provided
	Brand of adhesive	N/A	N/A	N/A	N/A	Not provided
	When adhesive applied	N/A	N/A	N/A	N/A	Not provided

Key Message :

- 100% In-line 2DID (SubMark) scanned is a general practices across all ODMs for traceability.
- Marking quality on 2DID is critical for flawless process execution.
- No adhesive process performed/capability; Pegatron is equipped with adhesive process but not performed on Intel component.

Others / Automation (L5/L6) - 1

Data Source:
Server ODMs
Inventec /
Foxconn TJ

Area		Quanta	Pegatron TW	Wistron ZS	Inventec / Foxconn TJ
Board Test	Partial / Full assembly process	Full automation	Partial: automated test machine (eg. DIMM, display card & etc) (suggest test automation development assessment align to NPI direction)	Not provided	Not provided
	In-line / off-line automation	In-line	in-line	in-line	
	Type of equipment/machines	Quanta in-house design	Semi-automated test machine (eg. Oceancat)	fixed auto test box	
	Pick & Place used ?	vacuum nozzle	Vacuum nozzle, gripper for CPU, and fixture for opening the CPU socket cover	NA	
Assembly Fixture	PHM assembly	No	No	N/A	
	Carrier to the CPU	No	yes	NA	

Key Message :

- Customers are exploring automated test at board functional test.
- ODMs' automation development is a continuous learning process, for improvement & enhancement. It is a trend / journey toward Industrial 4.0 (smart factory).

Others / Automation (L5/L6) - 2

Data Source:
Server ODMs

Area		Quanta	Pegatron TW	Wistron ZS	Inventec / Foxconn TJ
Assembly Time (Target)	Assembly of one PHM (CPU + carrier + heatsink)	30 - 40 s	~ 10 s	N/A	Not provided
	Attaching PHM assembly	< 60 s	~ 15 s	12 - 22 s	
	Throughput for standard 2S	< 120 s	50 s	N/A	
	Dissemble PHM from motherboard	< 120 s for 2S board (2S - 2 Socket solution)	20 s	N/A	
General	Any additional improvement required	*Socket PnP cover	No	*Socket cover need be designed for easy removal	
	Heatsink design	Follow Intel reference	Intel reference design, own customized design	Need a special heat-sink for mfg test (eg. Purley FTT)	
	Is the Intel PHM design automation friendly?	Not applicable, Designed PHM in Test fixture	Yes	No (PEEK nuts easily broken)	
	Torque Driver used, Brand and Model	generic brand	KILEWS SK-3120L KILEWS SK-8140L	Conos	

* Need details from customer

Key Message :

- EGS & beyond required automation friendly for retention assembly. Customers made request since Purley/2016.
- Retention design need to comprehend both manufacturing (automation) and field (manual) usage.
- Need continues support for assembly test tool development (Whitley → Eagle stream → future platform)

Others / Automation (L10) - 1

Data Source:
Server ODMs

Area		Quanta	Pegatron TW	Wistron ZS	Inventec / Foxconn TJ
System Test	Partial / Full assembly process	Partial * DIMM assembly * HDD Tray Assembly * Board screw assembly	No automation (suggest test automation development assessment align to NPI direction)	NA	Not provided
	In-line / off-line	Off-line	In-line	NA	
	Type of equipment/ machines are used?	Quanta in-house design and assembly	(suggest test automation development assessment align to NPI direction)	NA	
	Pick & Place used ?	Manual	Vacuum nozzle, gripper for CPU	NA	
Assembly Fixture	PHM assembly	No	No	*Based on Intel Design	
	Carrier to the CPU	No	yes	*Based on Intel Design	

Key Message :

- ODMs' automation development is a continuous learning process, for improvement & enhancement.
- Automation is a trend / journey toward Industrial 4.0 (smart factory).

Others / Automation (L10) - 2

Data Source:
Server ODMs

Area		Quanta	Pegatron TW	Wistron ZS	Inventec / Foxconn TJ
Assembly Time (Target)	Assembly of one PHM (CPU + carrier + heatsink)	N/A	~ 10 s	35 – 45 s	Not provided
	Attaching PHM assembly	< 60 s	~ 15 s	Not provided	
	Throughput for standard 2S	< 120 s	50 s	15 - 25 s	
	Dissemble PHM from motherboard	< 120 s for 2S board (2S – 2 socket)	20 s	5s for each screw	
General	Any additional improvement required	No	No	socket cover need be designed for easy removal	
	What heatsink design is used?	Follow Intel reference	Intel reference design, own customized design	N/A	
	Is the Intel PHM design automation friendly?	N/A Manual process to manage 1S/2S/4S variety	Yes	N/A	
	Torque Driver used, Brand and Model	generic brand	KILEWS SK-3120L KILEWS SK-8140L	Conos	

Key Message :

- Throughput time (TPT) target is varies by individual customer. Wistron showed lowest TPT.
- Higher manning ratio in Wistron to meet ~15s TPT for 2S PHM assembly (typical ~50s/operator).

Key Takeaway (1)

PCBA – Material / Consumables

- Stencil vendors preferable from local support for fast response, delivery time & on-site.
- Wave solder process is still required for thru-hole components.
- Shenmao paste/flux is commonly used except 1 customer.
 - Shenmao paste is generally used during early development; will be used as back-up during mass production (MP).
 - Alternate solder paste is mandatory for SMT per their customer requirement.
- Stepped stencil is commonly used, due to design complexity & large variety of component/pad size (varies in Area Ratio). Stepped up/down range between 3 – 10 mils.
- Stencil thickness for Intel component 4 – 5 mils.
- PCB thickness is within TMDG guideline.

PCBA – Process / Capability

- Wide variety of machine used and some reached maximum capability, need early development roadmap & engagement for tool capability assessment
 - PCB handling capability reached maximum width of 20 inch (Paste Print, SPI & AOI); 16 inch (PnP).
 - PnP (SMT & Rework) had reached maximum weight capability (~28g).
- Varies PCB support type used at paste print.
- Pallet usage is case/model dependent for paste print & PnP.

Key Takeaway (2)

PCBA – Process / Capability (continue)

- Customer's NPI typically followed MAS recommended profile. Profile will be customized profile in mass production (MP) to achieve good yield.
 - Peak Temp range of 20 °C due to large components/socket on PCB. Required range 235-255 °C, and 230-250 °C (customer depended)
 - TAL range required 60 – 90s and 60 -120s (customer dependent).
 - Delta_T <15°C (vs MAS <10°C).
- Rework @ air is still required.

Inspection Process (SPI / AOI / X-ray)

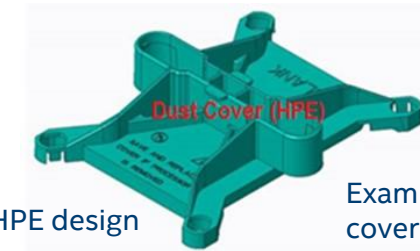
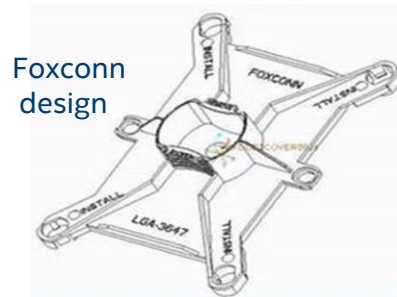
- 100% SPI & post-reflow AOI is performed during MP phase.
- Post-SMT X-ray sampling plan varies by customer.

Others / Automation

- 2DID (SubMark) is 100% scanned in-line for traceability. Marking output stability & quality is required to ensure stable 2DID reading.
- The traditional ODM manufacturing is transforming toward Industrial 4.0. Customer is exploring automation at board functional test process and other areas.
- Throughput time (TPT) target is varies by individual customer.

Next Step

- ❖ Conduct CRD survey for L10 factory in US region, to understand automation practice, requirement & trend.
- ❖ Need continues support for assembly test tool development (Whitley → Eagle stream → future platform). Possibly address customer concerns for breaking PEEK nuts issue.
- ❖ Seek customer clarification for PnP cover (new) and socket cover concerns (if this is existing/new issue).



DELL – available in eBay



Example of Purley socket cover improvement request

Rev Tracking

#	Rev	Date Released	Change Description
1	0	W26 Y2020	New doc
2	1	W32 Y2020	Updated machine information and (page 4 & 6, <i>in blue</i>)